Praveen Kumar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.57
Bhagat Public School, Kota, Rajasthan	Class XII (CBSE)	2024	90.20%
Jawahar Navodaya Vidyalaya, Agsauli, Hathras	Class X (CBSE)	2022	92.00%

SKILLS

Programming Languages: C++, Python, C.

Libraries: NumPy, Matplotlib, JAX.

Core CS: Data Structures & Algorithms, Object-Oriented Programming (OOP), Standard Template Library (STL), Mathematical Modelling.

Developer Tools: Git, GitHub, MATLAB, Solid Edge (Siemens).

Technical Areas: Computational Physics, Numerical Methods, Scientific Computing, Quantum Computing Fundamentals

MATLAB: Scientific plotting, numerical computation, ODE solving, Monte Carlo simulations and physics simulations.

CAD: Parametric CAD modelling and assembly design (Solid Edge).

PROJECTS

Statistical Arbitrage Framework Analysis (*Python, NumPy)*

- Designed a **market-neutral quantitative trading framework** leveraging **cointegration**, **mean reversion**, and **OLS regression** for statistical arbitrage.
- Formulated a trading pipeline involving **pair selection**, **Engle-Granger testing**, spread modelling, and **Z-score** based signal generation for systematic trade execution.
- Incorporated **beta-neutral** portfolio construction, **transaction cost modelling**, and **walk-forward validation** to improve strategy robustness and mitigate alpha decay.

Finite Difference Solver for One-Dimensional Hyperbolic PDEs (*Python, NumPy, Matplotlib*)

- Implemented a **Finite difference solver** in **Python** for one-dimensional hyperbolic PDEs using the **Lax-Friedrichs scheme**.
- Applied the **CFL stability condition** and implemented **zero-gradient boundary conditions** to ensure stable numerical evolution.
- Visualized the numerical solution using **NumPy** and **Matplotlib**, comparing wave propagation behavior across time to validate the implementation.
- Implemented **zero-gradient boundary conditions** using **ghost cells** to ensure stable numerical evolution and Simulated wave propagation over **2000 spatial grid points** to study numerical solution behavior.

JAX-Based Quantum Simulation Framework Analysis (*JAX, Python*)

Research Project

- Studied the mathematical foundations of **JAX-based quantum simulation**, including **Hilbert space representations**, **quantum operators**, and **time-evolution solvers**.
- Analyzed numerical formulations of the **Schrödinger**, **Lindblad Master**, and **Stochastic Schrödinger** equations used in differentiable quantum simulation.
- Evaluated **JAXQuantumSuite** example implementations against reference formulations to understand solver behavior and identify potential edge cases during simulation.

RELEVANT COURSES

Computer Science: Data Structures & Applications, Introduction to Programming, Computational Physics

Mathematics: Differential Equations, Linear Algebra, Probability, Mathematical Physics, Advanced Mathematical Physics

Physics: Quantum Mechanics-I, Optics & Photonics

COMPETITIVE PROGRAMMING

- Solved **435+** algorithmic problems across **Codeforces (176)**, **LeetCode (189)** and **GeeksforGeeks (70)**.
- Participated in **15** rated Codeforces contests.
- Secured **Global Rank 4227** in Codeforces Round 1021 (Div. 2).
- Earned **LeetCode 50 Days Badges** in **2025** and **2026**.
- Completed **220+** problems from the **Striver A2Z DSA Sheet** covering Arrays, Binary Search, Linked Lists, Trees, Graphs, Heaps and Dynamic Programming.

POSITIONS OF RESPONSIBILITY

Cultural Council Coordinator (Events)

- Coordinated event logistics and assisted in the execution of cultural activities under the Cultural Council.
- Collaborated with student coordinators to support planning and smooth conduct of institute cultural events.

EXTRACURRICULAR

Chess: Rapid rating **1295** (Peak: **1300**) on Chess.com.

National Service Scheme (NSS): Participated in community service and outreach initiatives.

Fitness: Consistent strength training focused on discipline and long-term physical development.

Additional Coursework: **Spoken Japanese Basics** (Grade: **A**)